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[Chap 6] Predictive Analytics and Learning Data Structures

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predictive analytics

- A method to predict invisible values based on observable data provide

- Most predict the future, but the subject may not necessarily be the future – predicting the future based on currently available data • e.g.) Future stock price prediction, demand forecasting
  - Predict a value that currently exists but is not visible to us
    - Example) Predicting the intrinsic value of a stock based on a company's financial data, etc.

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KNN (k-Nearest Neighbor) algorithm

- Also known as the k-nearest neighbor model

- Predict by following the majority of the k nearest neighbors

- Scikit-learn installation is required for the practice.
  - pip install sklearn (MacOS: pip3 install sklearn)

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Exploring the practice data (wine dataset)

- Load the wine dataset and print its contents

```
1  from sklearn import datasets
2  dataset = datasets.load_wine()
3  print(dataset)
```

– (Line 1) Import only the datasets module instead of the entire sklearn package

```
from (패키지명) import (모듈명)
```

- Print only the data descriptions of the wine dataset

```
1  from sklearn import datasets
2  dataset = datasets.load_wine()
3  print(dataset.DESCR)
```

– Data for a total of 178 wines – 13 characteristic

values for each wine • Alcohol content,

malic acid concentration, etc.

– Class (grade) of the wine: 3 types, 0, 1, and 2

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Exploring the Practice Data (Wine Dataset) (Continued)

- Outputting the types of attributes in the wine dataset

```
from sklearn import datasets
dataset = datasets.load_wine()
print(dir(dataset))
```

```
['DESCR', 'data', 'feature_names', 'target', 'target_names']
```

– DESCR: Dataset

description – data: 13 feature values for each of the 178 wines

- Output only data attributes

```
from sklearn import datasets
dataset = datasets.load_wine()
print(dataset.data)
```

```
[[1.423e+01 1.710e+00 2.430e+00 ... 1.040e+00 3.920e+00 1.065e+03]
 [1.320e+01 1.780e+00 2.140e+00 ... 1.050e+00 3.400e+00 1.050e+03]
 [1.316e+01 2.360e+00 2.670e+00 ... 1.030e+00 3.170e+00 1.185e+03]
 ...
```

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Exploring the Practice Data (Wine Dataset) (Continued)

- Take a closer look at data attributes

예제 ex6-1.py

```
1 from sklearn import datasets
2 dataset = datasets.load_wine()
3 print("data type=",type(dataset.data))
4 print("data shape=",dataset.data.shape)
5 print("first row of data=",dataset.data[0])
```

```
data type= <class 'numpy.ndarray'>
data shape= (178, 13)
first row of data= [1.423e+01 1.710e+00 2.430e+00 1.560e+01 1.270e+02 2.800e+00
3.060e+00 2.800e-01 2.290e+00 5.640e+00 1.040e+00 3.920e+00 1.065e+03]
```

– (Line 3) Print the data type: numpy multidimensional array – (Line 4)

Print the rows and columns of the data: 178 rows and 13

columns – (Line 5) Print the first row of the data (data[0]): The 13 characteristic values of the first wine are printed

- 1.423e+01 = 14.23 (exponential expression)

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Exploring the Practice Data (Wine Dataset) (Continued)

- Print feature names and correct answers

예제 ex6-2.py

```
1 from sklearn import datasets
2 dataset = datasets.load_wine()
3 print("feature names=",dataset.feature_names)
4 print("target=",dataset.target)
5 print("target names=",dataset.target_names)
```

– (Line 3) Output attribute name

```
feature names= ['alcohol', 'malic_acid', 'ash', 'alcalinity_of_ash', 'magnesium', 'total_
phenols', 'flavanoids', 'nonflavanoid_phenols', 'proanthocyanins', 'color_intensity',
'hue', 'od280/od315_of_diluted_wines', 'proline']
```

– (Row 4) Output correct answers for each of the 178 wines

```
target= [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2]
```

– (Line 5) Output the meaning of the correct answer value

```
target names= ['class_0' 'class_1' 'class_2']
```

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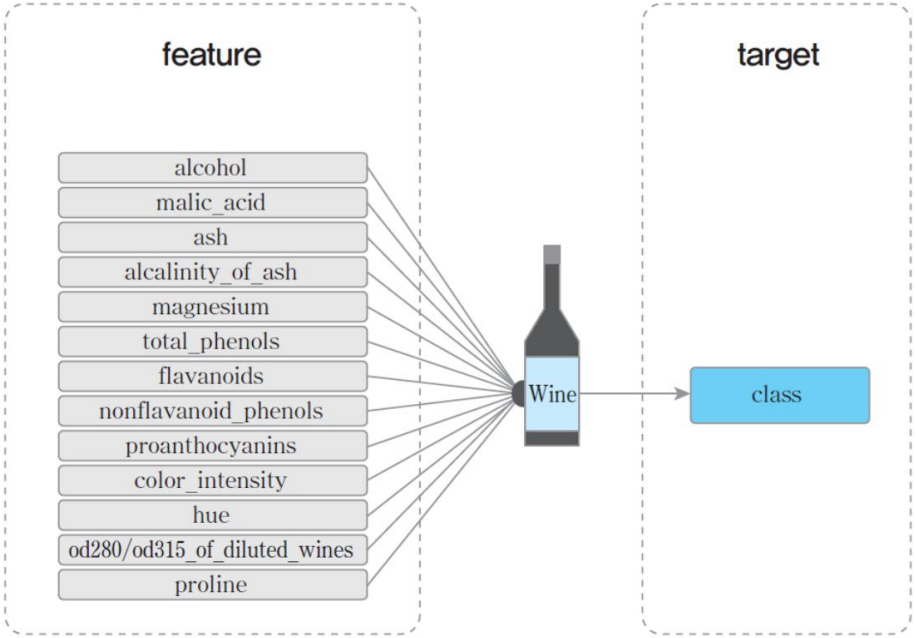
General structure of training datasets

- Features and Targets – Each wine
  has 13 feature values – The target for each
  wine's classification is indicated

- Data point – A single data point
  consisting of features and a ground truth
  – The wine dataset has 178 data points.

- Dataset: A collection of data points in large quantities



[그림 6–3] 와인 데이터의 구조

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2D visualization of wine data

- Transferring wine data to a pandas DataFrame

예제 ex6-4.py

```
1 import pandas as pd
2 from sklearn import datasets
3 dataset = datasets.load_wine()
4 d = pd.DataFrame(dataset.data, columns=dataset.feature_names)
5 d['target'] = dataset.target
6 print(d)
```

– (Row 4) Create a new DataFrame using dataset.data. Set the titles of each column to the names of the features appearing in the feature_names attribute.

– (Row 5) Add a new column named 'target' to the DataFrame created earlier, and set the value of datasets.target to that column.

	alcohol	malic_acid	ash	alcalinity_of_ash	magnesium	total_phenols	...	proanthocyanins	color_intensity	hue	od280/od315_of_diluted_wines	proline	target
0	14.23	1.71	2.43	15.6	127.0	2.80	...	2.29	5.64	1.04	3.92	1065.0	0
1	13.20	1.78	2.14	11.2	100.0	2.65	...	1.28	4.38	1.05	3.40	1050.0	0
2	13.16	2.36	2.67	18.6	101.0	2.80	...	2.81	5.68	1.03	3.17	1185.0	0
3	14.37	1.95	2.50	16.8	113.0	3.85	...	2.18	7.80	0.86	3.45	1480.0	0
4	13.24	2.59	2.87	21.0	118.0	2.80	...	1.82	4.32	1.04	2.93	735.0	0
..	...	...	...	...	...	...	...	...	...	...	...	...	...

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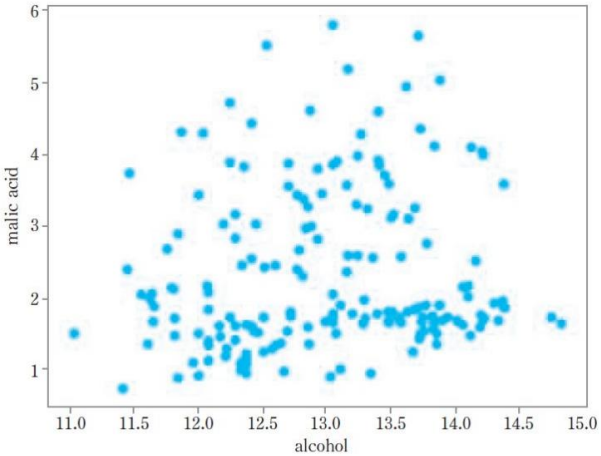
2D Visualization of Wine Data (Continued)

- Drawing a scatter plot with 'alcohol' and 'malic_acid' features

예제 ex6-5.py

```
1  import pandas as pd
2  from sklearn import datasets
3  dataset = datasets.load_wine()
4  d = pd.DataFrame(dataset.data, columns=dataset.feature_names)
5  d['target'] = dataset.target
6
7  import matplotlib.pyplot as plt
8  plt.scatter(d['alcohol'], d['malic_acid'])
9  plt.xlabel('alcohol')
10 plt.ylabel('malic_acid')
11 plt.show()
```



[그림 6-4] 와인 데이터의 alcohol과 malic_acid 산점도

– (Line 7) Load matplotlib.pyplot with the name plt – (Line 8)

Display scatter plots for each wine using d['alcohol'] on the x-axis and d['malic_acid'] on the y-axis

– (Lines 9–10) Display horizontal and vertical axis titles

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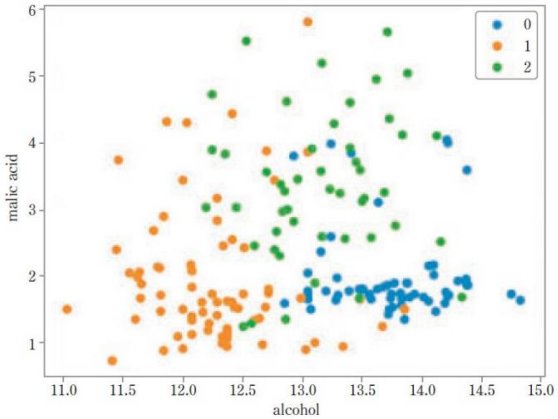
2D Visualization of Wine Data (Continued)

- Drawing a scatter plot by color-coding wine grades

예제 ex6-7.py

```
...
6  d = pd.DataFrame(dataset.data, columns=dataset.feature_names)
7  d['target'] = dataset.target
8  groups = d.groupby('target')
9  for group_name, group_data in groups:
10     plt.scatter(group_data['alcohol'], group_data['malic_acid'], label=group_name)
11 plt.legend()
12 plt.show()
```



[그림 6-5] 와인 등급별로 구분된 산점도

– (Line 8) Use the groupby command to divide into groups based on the value of target – (Line

9) Use a for loop to store each group name in group_name and the corresponding group dataframe in group_data – (Line 10)

Display the values of the 'alcohol' and 'malic_acid' columns of group_data for each group as the x-axis and y-axis, respectively, and set group_name as the label to apply different colors

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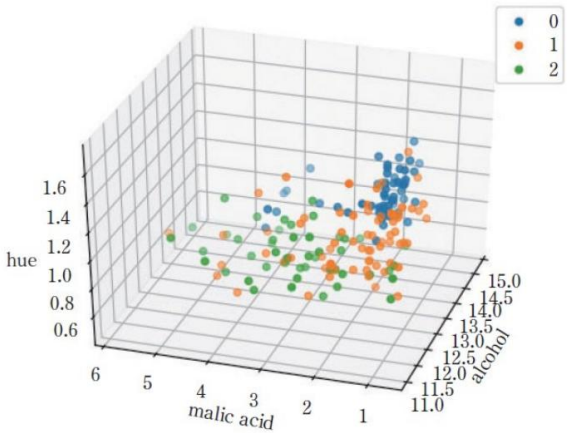
3D visualization of wine data

- 3D scatter plot with alcohol content on the x-axis, malic acid concentration on the y-axis, and color on the z-axis

예제 ex6-8.py

```
...
10 from mpl_toolkits.mplot3d import Axes3D
11 ax = plt.axes(projection = '3d')
12 for n, d in groups:
13     ax.scatter(d['alcohol'], d['malic_acid'], d['hue'], label=n)
14 plt.legend()
15 ax.set_xlabel('alcohol')
16 ax.set_ylabel('malic_acid')
17 ax.set_zlabel('hue')
18 plt.show()
```



[그림 6-6] 와인 데이터의 3차원 산점도

– (Line 10) Use the Axes3D module from the mpl_toolkits.mplot3d package for the 3D scatter plot – (Line 11) Set the coordinate axes in the projected 3D space – (Lines 12–13) Plot points for each grade using alcohol on the x-axis, malic_acid on the y-axis, and hue on the z-axis – By rotating the coordinate axes with the mouse, it can be confirmed that the areas where grades were mixed in the 2D scatter plot are actually significantly separated by the z-axis (color).

- Classification ability improves as additional characteristics are considered

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The concept of KNN

- Systematizing the idea that following the majority of neighbors will generally be correct
  - A wine with an alcohol content of 12 and malic acid of 2 is estimated to be

    Grade 1 • because wines that are almost entirely orange (Grade 1) are distributed around (12, 2).
  - Near (12.1, 3), there are many orange (Grade 1) points, but there are also green (Grade 2) points. •

    It is dangerous to judge blindly based on the nearest point. • It

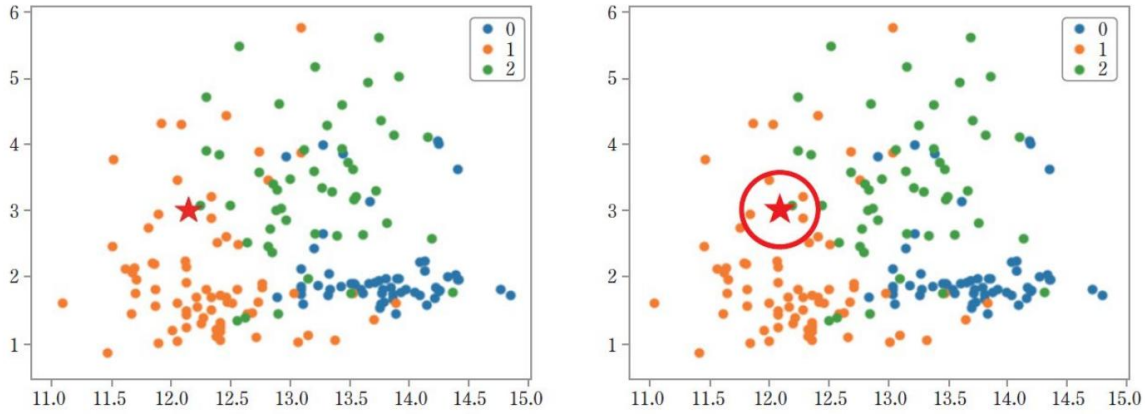
    would be safer to examine a few surrounding points and follow the general trend.

- KNN: A method of determining the type that occupies the majority among the k nearest points.
  - If k=5, use the five closest to (12.1, 3) as the basis.

    • Since the four nearest ones are orange and one is green, it is determined to be the majority orange (Grade 1).
  - Prediction performance is heavily influenced by the value

    of k. • k=1 is susceptible to being swayed by local peculiarities in the

    surroundings. • The key is to find the value of k that maximizes predictive power by varying it.



[그림 6-8] 도수 12.1, 사과산 3인 와인의 등급 판단

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Construction of KNN model

- Building training data

```
1  import pandas as pd
2  from sklearn import datasets
3  dataset = datasets.load_wine()
4  d = pd.DataFrame(dataset.data, columns=dataset.feature_names)
5  d['target'] = dataset.target
6
7  x_train = d[['alcohol','malic_acid']]
8  y_train = d['target']
9  print("<Feature data>")
10 print(x_train)
11 print("<Target data>")
12 print(y_train)
```

```
<Feature data>
     alcohol  malic_acid
0      14.23        1.71
1      13.20        1.78
2      13.16        2.36
..       ...         ...
176    13.17        2.59
177    14.13        4.10

[178 rows x 2 columns]
<Target data>
0      0
1      0
2      0
      ..
176    2
177    2
Name: target, Length: 178, dtype: int32
```

– (Line 7) Store only the 'alcohol' and 'malic_acid' columns from d in x_train

– (Line 8) Store only the 'target' column from d in y_train

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Building the KNN Model (Continued)

- Train the KNN model with data to predict

예제 ex6-9.py

```
...
14 from sklearn.neighbors import KNeighborsClassifier
15 knn = KNeighborsClassifier(n_neighbors=5)
16 knn.fit(x_train, y_train)
17 x_pred=pd.DataFrame(data = [[12, 3]], columns=['alcohol','malic_acid'])
18 print(x_pred)
19 print('Predicted :', knn.predict(x_pred))
```

```
(앞부분 생략)
   alcohol  malic_acid
0       12           3
Predicted : [1]
```

– (Line 14) Load the KNeighborsClassifier module for KNN from scikit-learn – (Line 15) Set n_neighbors corresponding to k values to 5 – (Line 16) Train using feature values x_train and ground truth values y_train – (Line 17) Predict for [12, 3]